

Subtracting Across a Zero

Answer Key

Grades 2–3

Quick Answers

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|---|---|
| 1. the difference = 163 | 6. the difference = 352 |
| 2. the difference = 333 | 7. the empty seats = 158 seats |
| 3. the difference = 445 | 8. (a) the correct difference = 264, — |
| 4. the difference = 224 | 9. the difference = 443 |
| 5. (a) the estimate = 400, (b) the exact difference = 427 | 10. (a) the estimate = 500, (b) the exact difference = 457, — |
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What is verified

8 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 2–3

3.NBT.A.2 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 14 tagged checks.

Common wrong answers

- 3. *If they got 555: took the smaller digit from the larger in every column instead of regrouping*
 - 4. *If they got 376: took the smaller digit from the larger in every column instead of regrouping*
 - 5. *If they got 500: rounded the number being taken away down instead of up*
 - 6. *If they got 448: took the smaller digit from the larger in every column instead of regrouping*
 - 8. *If they got 364: opened a hundred to feed the tens but left the hundreds digit unchanged*
 - 10. *If they got 460: rounded to the nearest ten instead of the nearest hundred*
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1. Subtract $305 - 142$.

Solution: Ones: $5 - 2 = 3$, no borrowing needed. Tens: zero cannot give four, so open one hundred into ten tens — the hundreds digit drops from three to two and the tens column becomes ten. Ten tens take away four tens leaves six. Hundreds: $2 - 1 = 1$. 163

2. Subtract $604 - 271$.

Solution: Ones: $4 - 1 = 3$. Tens: zero cannot give seven, so open a hundred — five hundreds left, ten tens in hand — and ten take away seven leaves three. Hundreds: $5 - 2 = 3$.

333

3. Subtract $703 - 258$.

Solution: This one borrows twice, and the order matters. The ones need help, but the tens column is empty, so first open a hundred: six hundreds and ten tens. Now the tens can lend one to the ones: nine tens and thirteen ones. Ones: thirteen take eight leaves five. Tens: nine take five leaves four. Hundreds: $6 - 2 = 4$.

445

4. Subtract $400 - 176$.

Solution: Both the ones and the tens are empty, so regroup once across the whole number: four hundreds becomes three hundreds and ten tens, then nine tens and ten ones. Ones: ten take six leaves four. Tens: nine take seven leaves two. Hundreds: $3 - 1 = 2$.

224

5. (a) Estimate $802 - 375$ to the nearest hundred. (b) Find the exact difference.

Solution (a): 802 rounds down to 800 and 375 rounds up to 400 (its tens digit is seven). Subtracting the rounded numbers gives $800 - 400$.

400

Solution (b): The ones need help and the tens are empty, so open a hundred first: seven hundreds, ten tens, then nine tens and twelve ones. Ones: twelve take five leaves seven. Tens: nine take seven leaves two. Hundreds: $7 - 3 = 4$. The exact answer sits close to the estimate, so nothing was dropped.

427

6. Subtract $600 - 248$.

Solution: Regroup across both empty columns: five hundreds, nine tens, ten ones. Ones: ten take eight leaves two. Tens: nine take four leaves five. Hundreds: $5 - 2 = 3$.

352

7. A theatre has 507 seats and 349 were taken. How many were empty?

Solution: “How many empty” is the whole minus the part taken, so subtract. The ones need help and the tens are empty: open a hundred, giving four hundreds and ten tens, then nine tens and seventeen ones. Ones: seventeen take nine leaves eight. Tens: nine take four leaves five. Hundreds: $4 - 3 = 1$.

158 seats

8. Rosa worked out $500 - 236$ and wrote 364. (a) Subtract correctly. (b) Explain what Rosa did wrong.

Solution (a): Regroup across the two zeros: four hundreds, nine tens, ten ones. Ones: ten take six leaves four. Tens: nine take three leaves six. Hundreds: $4 - 2 = 2$.

264

Solution (b) — instructor-judged. A correct response names the loan that was never paid. Full credit for the idea that a hundred was opened so the empty tens column had something to lend, but the hundreds column was then worked as five instead of the four that were left — so the answer came out a whole hundred too big. Accept an answer that writes the correct regrouping (four hundreds, nine tens, ten ones) and points at the hundreds column as the one that disagrees. Saying only that the answer is a hundred too big, without naming the column, earns half credit. Restating the correct difference earns no credit, since part (a) already asks for it.

9. Subtract $900 - 457$.

Solution: Regroup across both zeros: eight hundreds, nine tens, ten ones. Ones: ten take seven leaves three. Tens: nine take five leaves four. Hundreds: $8 - 4 = 4$. 443

10. (a) Estimate $806 - 349$ to the nearest hundred. (b) Find the exact difference. (c) Explain how the estimate shows the answer is reasonable.

Solution (a): 806 rounds down to 800 and 349 rounds down to 300 (its tens digit is four). The estimate is $800 - 300$. 500

Solution (b): Ones need help, tens are empty: open a hundred, giving seven hundreds and ten tens, then nine tens and sixteen ones. Ones: sixteen take nine leaves seven. Tens: nine take four leaves five. Hundreds: $7 - 3 = 4$. 457

Solution (c) — instructor-judged. A correct response uses the estimate as a test, not as proof. Full credit for noting that the exact difference lands within about fifty of the estimate and explaining the direction: the number being taken away was rounded down, so the estimate takes away less and must sit above the exact answer. Accept the check stated the other way round — an answer differing from the estimate by a whole hundred would be the signature of a lost regrouping. Comparing the two numbers with no reason earns half credit. Saying the estimate proves the exact answer correct earns no credit.